

LIFT-SE: LINGUISTIC INFERENCE FOLLOWED BY FLOW TRANSFORMATION FOR GENERATIVE SPEECH ENHANCEMENT

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ABSTRACT

Generative speech enhancement (SE) is prone to linguistic hallucination when semantic constraint is unreliable under severe noise and reverberation. Moreover, approaches that generate discrete codec tokens are bounded by the quantization error of the codec decoder, regardless of token-prediction accuracy. We propose LIFT-SE, a two-stage generative framework that decouples linguistic inference from acoustic synthesis within QRes-Codec, which exposes a quantized latent and its residual-completed continuous form. The first stage predicts clean codec tokens autoregressively, conditioned on frame-aligned features from a self-supervised front-end distilled toward clean speech. The second stage applies conditional flow matching to transport Gaussian noise to the continuous latent conditioned on the predicted tokens, and the frozen decoder reconstructs the enhanced waveform. Discrete generation provides naturalness, while continuous refinement restores signal fidelity. Experiments on the DNS1 and URGENT benchmarks show that LIFT-SE attains favorable linguistic consistency under reverberant conditions together with competitive perceptual quality, and systematic ablations verify the necessity of both stages. Code will be released in the future.

Index Terms— Speech enhancement, autoregressive language model, conditional flow matching, neural audio codec, self-supervised feature distillation

1. INTRODUCTION

Speech enhancement (SE) aims to recover clean speech from noisy mixtures, improving perceptual quality and intelligibility for both human listeners and downstream applications such as automatic speech recognition (ASR) [1] and speaker verification [2]. Deep-learning-based SE methods can be broadly grouped into discriminative approaches [3, 4], which estimate clean speech or its representations from noisy input, and generative approaches, which synthesize enhanced speech from learned speech priors. The latter include diffusion models [5, 6], flow matching models [7, 8], and language-model (LM)-based approaches [9, 10, 11]. Generative SE can produce highly natural-sounding speech, but this capability entails a

fundamental risk: perceptual plausibility does not guarantee content faithfulness, i.e., linguistic hallucination [12, 13].

Achieving content faithfulness becomes particularly challenging when noise or reverberation obscures the acoustic evidence for the spoken words. Many LM-based SE systems generate discrete speech tokens, such as codec tokens, conditioned on representations of the degraded signal [14, 11]. When the conditioning is corrupted by severe noise or reverberation, the autoregressive prior dominates and the model produces fluent yet linguistically incorrect content [15]. Continuous generative models similarly drift toward plausible but unfaithful speech in the absence of an explicit semantic anchor [7, 5]. Existing work strengthens semantic guidance through noise-invariant representations: L3-SE distills noisy-speech representations toward clean acoustic and semantic teachers and reports strong linguistic consistency under reverberant conditions [15]; PASE and StuPASE instead constrain generation in the continuous domain with a distilled self-supervised-learning (SSL) phonological prior [13, 8]. However, the discrete modeling of L3-SE leaves reconstruction quality bounded by the quantization loss of the codec decoder, whereas the continuous-representation modeling of PASE and StuPASE falls short of autoregressive generation in naturalness. SenSE combines discrete token generation with flow-based waveform synthesis, but its token predictions and flow targets use different representations, requiring a mapping between the two stages [16]. These observations motivate a framework that performs discrete linguistic inference and continuous acoustic refinement over a shared underlying speech representation.

We therefore propose LIFT-SE, a two-stage generative SE framework that decouples linguistic inference from acoustic synthesis. LIFT-SE is built on QRes-Codec, which exposes a quantized latent z_q for discrete prediction and its residual-completed continuous form $z_q + z_{\text{res}}$ for high-fidelity reconstruction. Concretely, the first stage is a conditional autoregressive (AR) model that predicts clean discrete codec tokens, conditioned on frame-aligned features from an SSL front-end and a frozen codec encoder that are injected as prepended key-value pairs at every decoder layer. The SSL front-end is adapted with low-rank adaptation (LoRA) and distilled toward the clean-speech representations of the same frozen backbone, which thus serves as its own teacher. The second stage is a

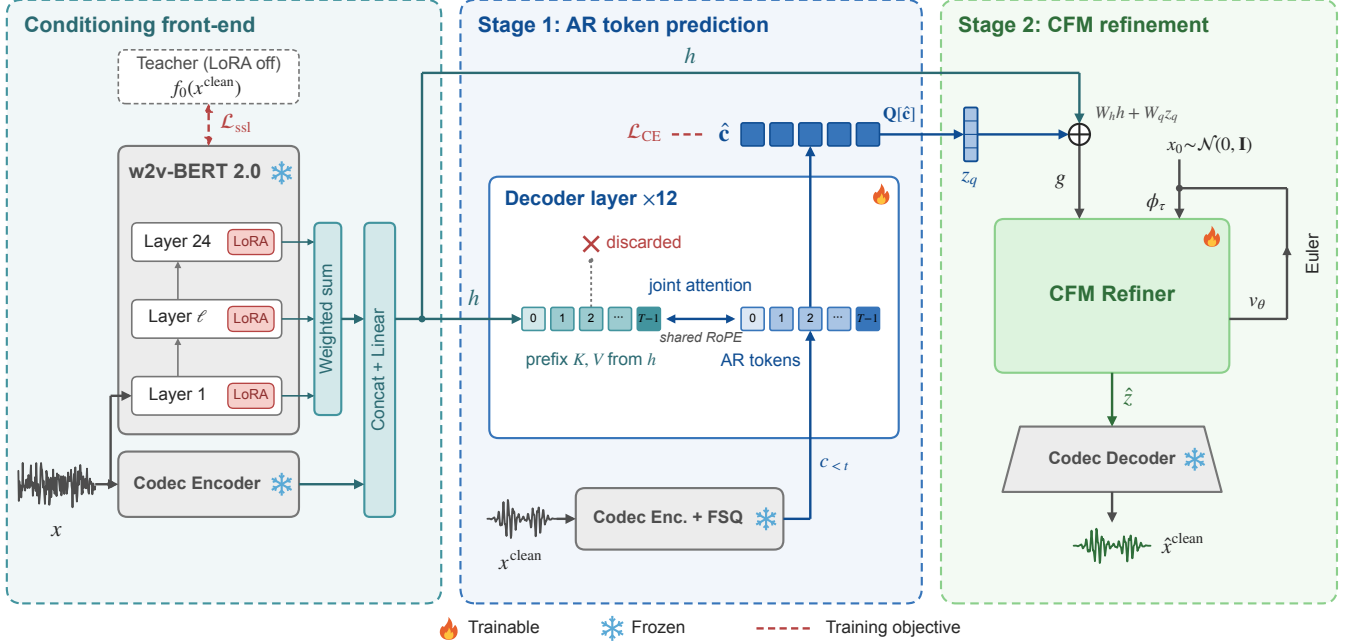


Fig. 1. Overview of LIFT-SE. A LoRA-adapted self-supervised front-end and a frozen QRes-Codec encoder form the frame-aligned conditioning. Stage 1 predicts clean codec tokens, and Stage 2 refines their quantized latents into continuous latents before waveform reconstruction by the frozen QRes-Codec decoder.

conditional flow matching (CFM) refiner [17] that transports Gaussian noise to the continuous latent of QRes-Codec, conditioned on the AR-predicted tokens; the frozen codec decoder then reconstructs the enhanced waveform.

Our contributions are as follows. First, we present QRes-Codec, in which discrete prediction and continuous refinement share a single representation space, eliminating the cross-representation conversion required when tokens and flow targets come from different representations. Second, we design a two-stage framework in which discrete autoregressive generation provides naturalness and an explicit causal decision over linguistic content, while continuous refinement recovers the information that quantization discards, lifting restoration quality beyond the fidelity of the discrete path. Third, on the DNS1 and URGENT benchmarks, LIFT-SE compares favorably with strong generative baselines under reverberant conditions while maintaining competitive perceptual quality. Removing either stage degrades the result, indicating that the two stages are complementary.

2. PROPOSED METHOD

2.1. Overview

Figure 1 presents LIFT-SE, which enhances a noisy waveform x in two stages that share a frozen QRes-Codec. QRes-Codec represents speech at a frame rate of 50 Hz as a discrete token sequence $\mathbf{c} = (c_1, \dots, c_T)$ and a continuous latent sequence.

Let $\mathbf{Q} \in \mathbb{R}^{K \times 32}$ denote the dequantization table, where $K = 4096$ is the codebook size. A table lookup maps tokens to quantized latents, $z_q = \mathbf{Q}[\mathbf{c}]$, allowing both stages to operate on representations defined by the same codec.

Stage 1 predicts clean codec tokens autoregressively from the noisy input and obtains z_q by table lookup. Stage 2 uses conditional flow matching to estimate the target continuous latent $z = z_q + z_{\text{res}}$, where z_{res} restores information omitted by quantization. The frozen codec decoder converts the refined latent $\hat{z}$ into the enhanced waveform $\hat{x}$. QRes-Codec and the self-supervised backbone (w2v-BERT-2.0) [18] remain frozen, whereas the LoRA adapters, AR model, and CFM refiner are trained.

2.2. QRes-Codec with semantic distillation

Inspired by the homogeneous Transformer autoencoder design of MOSS-Audio-Tokenizer [19], QRes-Codec (Figure 2) processes 16-kHz waveforms using non-causal Transformer blocks. The waveform is divided into 320-sample patches, yielding a frame rate of 50 Hz; both the encoder and decoder contain 20 Transformer layers with a hidden dimension of 768. Two projection heads each map the encoder output to a 32-dimensional latent. The first head produces z_{vq} , which is quantized with finite scalar quantization (FSQ) [20] using levels $[4, 4, 4, 4, 4, 4]$. The resulting single codebook contains $4^6 = 4096$ entries and has a bit rate of 600 bps. The second head produces the residual z_{res} . A single decoder reconstructs

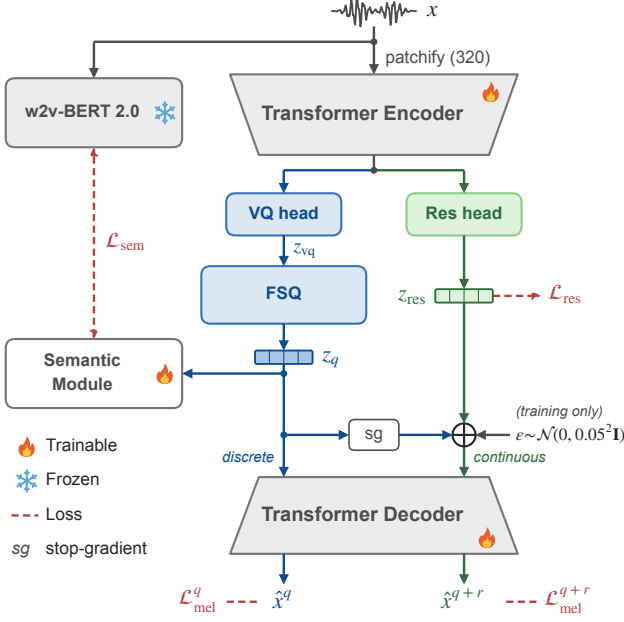


Fig. 2. The frozen QRes-Codec provides a quantized latent z_q for discrete prediction and a continuous latent $z_q + z_{\text{res}}$ for high-fidelity reconstruction. Semantic distillation is applied to the quantized branch.

speech from z_q in the discrete path and from $z_q + z_{\text{res}}$ in the continuous path.

QRes-Codec is trained independently and then frozen for speech enhancement. Its objective combines reconstruction, residual, semantic, adversarial, and feature-matching terms:

$$\mathcal{L}_{\text{codec}} = 10\mathcal{L}_{\text{mel}}^q + 10\mathcal{L}_{\text{mel}}^{q+r} + 10\mathcal{L}_{\text{res}} + 5\mathcal{L}_{\text{sem}} + \mathcal{L}_{\text{adv}} + \mathcal{L}_{\text{fm}}, \quad (1)$$

where $\mathcal{L}_{\text{mel}}^q$ and $\mathcal{L}_{\text{mel}}^{q+r}$ denote multi-resolution log-mel losses for the discrete and continuous paths, respectively. During training, the quantized latent is detached in the continuous path, and $\mathcal{L}_{\text{res}} = \|z_{\text{res}}\|_2^2$ encourages the codes to retain the principal information. Multi-period and multi-resolution discriminators provide $\mathcal{L}_{\text{adv}}$ and $\mathcal{L}_{\text{fm}}$. Following semantic codec training [21], a lightweight semantic module built on ConvNeXt [22] maps z_q to the layer-17 representation of a frozen w2v-BERT-2.0 model [18]; the mean-squared error defines $\mathcal{L}_{\text{sem}}$. This loss encourages the quantized representation to retain linguistic information. Gaussian noise with a standard deviation of 0.05 is added only to $z_q + z_{\text{res}}$ during codec training, improving the robustness of the downstream refiner.

2.3. Stage 1: conditional autoregressive token prediction

Conditioning front-end. We extract features from noisy speech x using w2v-BERT-2.0, whose pretrained weights are frozen. Each of its 25 hidden states is normalized by its own

precomputed root-mean-square constant, and the normalized states are combined using learned softmax weights. The result is concatenated with the frame-aligned representation from the frozen codec encoder and projected to a sequence of fusion tokens $h \in \mathbb{R}^{T \times 1024}$. Rank-16 LoRA adapters are added to the query, key, value, and output projections of each w2v-BERT-2.0 attention layer.

The LoRA adapters learn noise-invariant conditioning by aligning noisy-speech features with the clean-speech features of the same frozen backbone, following the noisy-to-clean representation distillation in PASE [13]. Let f_L denote the backbone with LoRA enabled and f_0 the backbone with LoRA disabled. The latter is evaluated without gradients and serves as the teacher:

$$\mathcal{L}_{\text{ssl}} = \|\text{LN}(f_L(x)) - \text{LN}(f_0(x^{\text{clean}}))\|_2^2. \quad (2)$$

Here, $\text{LN}(\cdot)$ denotes layer normalization. Thus, the teacher exists but requires no additional training.

Autoregressive model. The AR model is a decoder-only Transformer that follows the Qwen2 architecture [23]. It contains 12 layers with a hidden dimension of 768, 12 attention heads, an intermediate dimension of 3072, and a vocabulary of 4096 codec tokens. No end-of-sequence token is needed because the output length equals the conditioning length. The model factorizes the clean token sequence as

$$p_{\theta}(c | h) = \prod_{t=1}^T p_{\theta}(c_t | c_{<t}, h). \quad (3)$$

Frame-aligned conditioning. At each decoder layer, an independent projection maps the fusion tokens h to conditional key and value vectors, which are prepended to the keys and values derived from the AR-token states. After attention, outputs at the fusion-token positions are discarded, and only the outputs at the AR-token positions are propagated to the next layer. The fusion tokens and AR tokens share rotary indices $0, \dots, T-1$; consequently, the two tokens at the same frame have zero relative rotary distance, independent of T . This framewise alignment allows inference to accommodate input audio of arbitrary duration.

Objective and inference. Stage 1 is trained with teacher forcing under

$$\mathcal{L}_1 = \mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{ssl}}, \quad (4)$$

where $\mathcal{L}_{\text{CE}}$ is the token cross-entropy. At inference, greedy generation produces T tokens and the dequantization table maps them to z_q .

2.4. Stage 2: conditional flow matching refinement

Stage 2 refines z_q in the continuous codec representation rather than in a mel-spectrogram space. The 32-dimensional codec latent is lower-dimensional than typical mel-spectrogram representations, and it can be decoded directly without a separately

trained vocoder. Let $s_1 = z = z_q^{\text{clean}} + z_{\text{res}}^{\text{clean}}$ be the target continuous latent, where the superscript *clean* marks latents obtained from the corresponding clean speech, and let $s_0 \sim \mathcal{N}(0, I)$ be the noise endpoint. For flow time $\tau \sim \mathcal{U}(0, 1)$, we use the optimal-transport path [17]

$$\phi_\tau = (1 - \tau)s_0 + \tau s_1, \quad u_\tau = s_1 - s_0, \quad (5)$$

and train the velocity field with

$$\mathcal{L}_2 = \mathbb{E}_{s_0, s_1, \tau} \left[\|v_\theta(\phi_\tau, \tau \mid h, z_q) - u_\tau\|_2^2 \right]. \quad (6)$$

Here, z_q is obtained from Stage 1, and the fusion tokens h provide the noisy-speech evidence used by both stages.

The refiner consists of eight bidirectional Transformer blocks with a hidden dimension of 512, eight attention heads, and an intermediate dimension of 2048. Its condition is

$$g = W_h h + W_q z_q. \quad (7)$$

The condition g and $W_x \phi_\tau$ are concatenated along the time axis, and the two segments use the same frame indices for rotary position encoding. Here, W_h , W_q , and W_x are learnable linear projections that map the fusion tokens h , quantized latent z_q , and interpolated latent ϕ_τ , respectively, to the refiner hidden dimension. The velocity prediction head is applied only to the outputs corresponding to the latent segment. A sinusoidal embedding of τ modulates every Transformer block through adaptive layer normalization [24]. During training, the entire condition is dropped with probability 0.2 to enable classifier-free guidance [25].

At inference, Stage 1 first generates $\hat{c}$ greedily, and the dequantization table yields $z_q = \mathbf{Q}[\hat{c}]$. Starting from $s^{(0)} = s_0$, Euler integration updates

$$s^{(i+1)} = s^{(i)} + \Delta\tau_i [v_c + \gamma(v_c - v_u)], \quad (8)$$

where v_c and v_u are the velocities predicted with and without the condition, respectively, γ is the guidance strength, and $\Delta\tau_i$ is the step size of the i -th integration step. The final latent $\hat{z}$ is passed to the frozen codec decoder to obtain $\hat{x}$.

3. EXPERIMENTS

3.1. Datasets

QRes-Codec is trained on VoxBox [26], which contains approximately 100,000 hours of speech. For SE training, clean speech is drawn from Multilingual LibriSpeech [27] and Emilia [28], about 3,000 hours in total, and the noise collection, about 330 hours in total, comprises the DNS Challenge noise set [29], DESED [30], DISCO [31], WHAM! [32], FSD50K [33], ESC-50 [34], DEMAND [35], MUSAN [36], and TUT Urban Acoustic Scenes 2018 [37]. All training mixtures are generated online at 16 kHz. We apply a simulated room impulse response (RIR) from SLR28 with

probability 0.5; the corresponding target is generated with the same RIR truncated 1 ms after its direct-path peak. Noise is added with probability 0.8 at an SNR sampled uniformly from $[-5, 15]$ dB, after which the mixture is normalized to a level sampled from $[-30, -15]$ dB. We add three further distortions to the simulation to obtain a universal enhancement version, LIFT-SE-U: bandwidth limitation with probability 0.2, which band-limits the signal to a sampling rate drawn from $\{3, 4, 6, 8, 11.025, 12, 16\}$ kHz; clipping with probability 0.2, with the lower and upper thresholds set at sample quantiles drawn from $[0, 0.1]$ and $[0.9, 1]$; and packet loss with probability 0.15, which drops 20 ms packets at a loss rate drawn from $[0.05, 0.25]$.

QRes-Codec is evaluated on the full LibriSpeech *test-clean* split [38] (2620 utterances) as a clean-speech reconstruction task without simulated degradation. SE is evaluated on the synthetic DNS1 test set [29], which contains a *no-reverb* and a *with-reverb* subset of 150 utterances each. For *with-reverb*, the reference is the anechoic clean speech, so the evaluation covers dereverberation as well as denoising. LIFT-SE-U is additionally evaluated on the non-blind test set of the Interspeech 2025 URGENT challenge [39], which contains 1000 multilingual utterances degraded by up to five of seven distortion types: additive noise, reverberation, clipping, bandwidth limitation, codec distortion, packet loss, and wind noise. The input sampling rates range from 8 to 48 kHz and are uniformly resampled to 16 kHz for processing.

3.2. Baselines

We compare QRes-Codec with EnCodec [40], DAC [41], X-Codec [42], WavTokenizer [43], UniCodec [44], X-Codec2 [45], and BiCodec [26]. For SE, we include representative discriminative, continuous-domain, discrete-token, and hybrid systems. TF-GridNet [3] is evaluated using the URGENT 2025 baseline checkpoint [39]. PASE [13], UniPASE [46], and StuPASE [8] use distilled self-supervised representations, with StuPASE performing CFM generation. UniSE [11] predicts discrete codec tokens autoregressively, and SenSE [16] combines discrete semantic tokens with flow matching. We run all systems from their official checkpoints and apply the same evaluation pipeline to their outputs.

3.3. Evaluation metrics

We assess the outputs from four perspectives.

- **Perceptual quality:** DNSMOS P.835 [47] and UTMOS [48], which predict perceived quality without a reference. We report the overall (OURL) score of DNSMOS; because DNSMOS is sensitive to signal level, each enhanced output is scaled to the root-mean-square level of its noisy input before scoring.
- **Signal fidelity:** wideband PESQ [49] and STOI [50], which compare the output with the reference waveform.

Table 1. QRes-Codec reconstruction on LibriSpeech *test-clean*. FPS denotes frames per second, N_q the number of codebooks, and BPS the bit rate in bits per second. Bold marks the best result excluding ground truth.

Model	FPS	N_q	BPS	PESQ $\uparrow$	STOI $\uparrow$	UTMOS $\uparrow$	SIM $\uparrow$	WER $\downarrow$
Ground truth	–	–	–	4.64	1.00	4.09	1.00	2.43
EnCodec	75	8	6000	2.77	0.94	3.09	0.89	2.64
DAC	50	4	2000	1.42	0.84	1.83	0.60	4.32
X-Codec	50	2	1000	2.33	0.86	4.21	0.72	3.13
WavTokenizer	75	1	900	2.39	0.91	4.00	0.68	5.43
UniCodec	75	1	900	2.56	0.92	4.00	0.76	4.23
X-Codec2	50	1	800	2.43	0.92	4.13	0.82	3.53
BiCodec	50	1	650	2.51	0.92	4.18	0.80	3.23
QRes-Codec (z_q)	50	1	600	2.22	0.91	4.19	0.69	5.64
QRes-Codec ($z_q + z_{res}$)	50	–	–	3.75	0.97	4.22	0.92	2.72

- **Linguistic consistency:** SpeechBERTScore (SBS) [51], Levenshtein phoneme similarity (LPS) [12], and word error rate, where SBS and LPS follow the URGENT 2025 challenge implementation [39]. For QRes-Codec, WER is computed against the ground-truth text with a HuBERT-Large model.¹ DNS1 provides no transcripts, so for SE we report the differential WER (dWER) between the Whisper large-v3 [52] transcripts of the enhanced speech and of the clean reference. For URGENT 2025, we report the WER between the transcripts of the output speech and the reference text.
- **Speaker preservation:** speaker similarity (SIM), the cosine similarity between speaker embeddings of the output and the reference, extracted by a speaker verification model based on WavLM-Large features [53].²

For QRes-Codec, we report PESQ, STOI, UTMOS, SIM, and WER; for SE, we report DNSMOS, UTMOS, SBS, LPS, SIM, dWER, and WER. WER and dWER are reported as percentages. Higher is better for all metrics except WER and dWER.

3.4. Implementation details

QRes-Codec contains approximately 307M parameters and is trained with AdamW using a learning rate of 1.8×10^{-4} for both the generator and the discriminators and gradient clipping at 5.0; it remains frozen in all SE experiments. The 580M-parameter w2v-BERT 2.0 backbone is also frozen, while rank-16 LoRA adapters with $\alpha = 32$ and dropout 0.05 are applied to the query, key, value, and output projections of every attention layer, adding approximately 3.1M trainable parameters. The 140M-parameter AR model is trained for 80k updates using AdamW with a peak learning rate of 3×10^{-4} . The 49.5M-parameter CFM refiner is trained for 50k updates with the same optimizer settings while Stage 1 is frozen, with token

¹<https://huggingface.co/facebook/hubert-large-ls960-ft>

²https://github.com/microsoft/UniSpeech/tree/main/downstreams/speaker_verification

conditions sampled from teacher-forced Stage 1 predictions. At inference, Stage 1 decodes greedily, and the refiner uses 4 Euler steps (6 for LIFT-SE-U) on a sway-sampled time grid with coefficient -1 [54] and guidance strength $\gamma = 0.5$. LIFT-SE contains approximately 193M trainable parameters.

3.5. QRes-Codec evaluation

Table 1 compares the discrete and continuous reconstruction paths of QRes-Codec. At 600 bps, the quantized path obtains a UTMOS of 4.19, but its PESQ, SIM, and WER indicate a substantial loss of fine acoustic detail, speaker characteristics, and linguistic content. Since UTMOS is a learned non-intrusive predictor, values above that of the ground-truth recordings should not be interpreted as higher reconstruction fidelity [55].

The continuous path retains similar predicted naturalness while improving PESQ, STOI, SIM, and WER to 3.75, 0.97, 0.92, and 2.72, respectively, indicating that z_{res} contains acoustic information critical to improving fidelity. The contrast between the two paths motivates the proposed decomposition: Stage 1 predicts the discrete representation, and Stage 2 recovers the information absent from that representation.

3.6. Speech enhancement results

Table 2 compares the systems on DNS1. On *no-reverb*, LIFT-SE performs competitively in predicted naturalness and phonetic preservation. Under reverberant conditions, LIFT-SE attains the lowest dWER together with the highest UTMOS and speaker similarity, exhibiting a favorable balance across naturalness, speaker similarity, and linguistic fidelity. In contrast, the high DNSMOS and UTMOS of UniSE do not translate into comparable dWER, underscoring the need to assess generative SE with both perceptual and content-based metrics. LIFT-SE-U performs comparably to LIFT-SE on both subsets, indicating that training on a wider range of distortions does not degrade performance in conventional scenarios. Spectrograms and transcripts of selected *with-reverb* utterances are provided in Appendix A.

3.7. Evaluation on URGENT 2025

We evaluate LIFT-SE-U on the URGENT 2025 non-blind test set to assess its performance on universal speech enhancement. As shown in table 3, all enhancement systems raise DNSMOS and UTMOS well above the noisy input. UniSE obtains the highest DNSMOS, but its WER is very high, whereas TF-GridNet and UniPASE keep the WER close to that of the noisy input. Although LIFT-SE-U sees no codec distortion or wind noise during training, it still obtains the best UTMOS, WER, and LPS. However, its SIM is lower than that of TF-GridNet, a purely discriminative model, possibly because generative modeling tends to restore natural-sounding speech under more complex distortions at the cost of some speaker characteristics.

Table 2. Speech enhancement results on the DNS1 test set. Bold marks the best result.

Model	DNS1 <i>no-reverb</i>						DNS1 <i>with-reverb</i>					
	DNSMOS↑	UTMOS↑	SBS↑	LPS↑	SIM↑	dWER↓	DNSMOS↑	UTMOS↑	SBS↑	LPS↑	SIM↑	dWER↓
Noisy	2.48	2.36	0.88	0.90	0.96	3.81	1.39	1.30	0.74	0.63	0.79	10.99
TF-GridNet	3.30	3.88	0.94	0.97	0.96	3.61	2.60	1.41	0.83	0.88	0.80	9.35
PASE	3.38	4.02	0.94	0.97	0.92	2.99	2.70	1.64	0.86	0.90	0.75	10.27
StuPASE	3.39	3.92	0.94	0.97	0.88	2.48	3.37	3.91	0.89	0.92	0.70	8.88
UniPASE	3.38	4.04	0.95	0.97	0.95	2.71	3.32	3.61	0.90	0.93	0.76	9.00
SenSE	3.39	3.85	0.94	0.97	0.92	5.78	3.38	3.56	0.88	0.92	0.76	11.72
UniSE	3.42	4.09	0.91	0.93	0.78	8.64	3.41	3.78	0.80	0.74	0.75	35.56
LIFT-SE	3.35	4.05	0.95	0.97	0.91	3.35	3.40	4.15	0.90	0.93	0.81	8.85
LIFT-SE-U	3.37	4.07	0.95	0.97	0.92	4.00	3.40	4.12	0.90	0.93	0.80	8.94

Table 3. Results on the URGENT 2025 non-blind test set. Bold marks the best result among the enhancement systems.

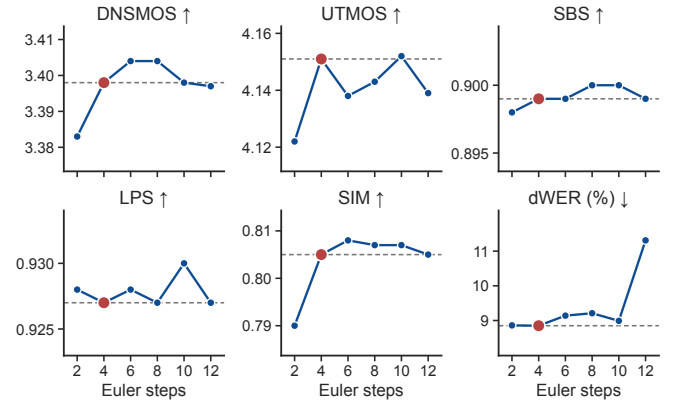
Model	DNSMOS↑	UTMOS↑	SBS↑	LPS↑	SIM↑	WER↓
Noisy	1.83	1.56	0.75	0.62	0.81	27.99
TF-GridNet	2.96	2.13	0.86	0.79	0.86	27.67
UniPASE	3.21	3.00	0.89	0.84	0.83	27.78
UniSE	3.26	3.08	0.78	0.60	0.58	54.90
LIFT-SE-U	3.24	3.22	0.87	0.84	0.81	23.11

3.8. Ablation study

Table 4 isolates the design choices of each stage. Comparing Stage 1-only decoding with LIFT-SE shows that continuous refinement improves speaker similarity and linguistic fidelity while retaining high predicted naturalness, supporting the intended division of labor between the two stages.

Stage 1 conditioning. Removing the SSL features causes the largest degradation across linguistic, phonetic, and speaker-related metrics, particularly under reverberation. Removing LoRA also substantially weakens linguistic consistency, whereas removing the distillation loss and the codec-encoder features produces a smaller but consistent decline. These trends show that the SSL representation provides the main linguistic constraint and that noisy-to-clean adaptation makes this constraint more robust to reverberation.

Refiner. The regression refiner obtains higher DNSMOS and UTMOS, whereas CFM yields higher speaker similarity and a lower dWER on both subsets. Removing the z_q condition raises the dWER, reflecting the contribution of Stage 1 to linguistic fidelity. Training the refiner on generated tokens, i.e., on tokens that Stage 1 generates autoregressively as at inference instead of teacher-forced predictions, yields the lowest *with-reverb* dWER among the refiner variants, accompanied by slight reductions in predicted perceptual quality. This result suggests that exposure to Stage 1 generation errors may help the refiner correct some of these errors under reverberation at the cost of slightly reduced naturalness.

**Fig. 3.** Effect of the number of Euler steps on DNS1 *with-reverb* for LIFT-SE. The red marker and the dashed line denote the default four steps.

Number of flow steps. As shown in Figure 3, all metrics change little with the number of Euler steps. Therefore, we utilize four steps to provide a good balance between quality and inference cost.

4. CONCLUSION

We presented LIFT-SE, a two-stage generative SE system that separates discrete linguistic inference from continuous acoustic refinement within the representation space of QRes-Codec. Evaluations of QRes-Codec and LIFT-SE consistently show that autoregressive token prediction primarily establishes speech naturalness, while refinement in the continuous latent improves signal fidelity, speaker preservation, and linguistic consistency. On DNS1, LIFT-SE is particularly competitive under reverberation, and the ablations support the complementary roles of the two stages. Extending the training simulation to more distortion types largely maintains this behavior on DNS1 and yields a favorable word error rate on the multi-distortion URGENT 2025 non-blind test set. Some limitations remain:

Table 4. Ablations on DNS1. Stage 1 variants are decoded directly from z_q without the refiner and are compared with the Stage 1-only system; refiner variants are compared with LIFT-SE. Bold marks the best result.

Variant	DNS1 <i>no-reverb</i>						DNS1 <i>with-reverb</i>					
	DNSMOS↑	UTMOS↑	SBS↑	LPS↑	SIM↑	dWER↓	DNSMOS↑	UTMOS↑	SBS↑	LPS↑	SIM↑	dWER↓
LIFT-SE (CFM refiner)	3.35	4.05	0.95	0.97	0.91	3.35	3.40	4.15	0.90	0.93	0.81	8.85
Stage 1 only	3.42	4.19	0.92	0.94	0.73	6.72	3.40	4.20	0.88	0.91	0.67	11.28
w/o $\mathcal{L}_{ssl}$	3.43	4.19	0.92	0.93	0.73	7.77	3.40	4.16	0.88	0.91	0.66	12.21
w/o LoRA	3.42	4.19	0.91	0.93	0.72	8.34	3.38	4.09	0.85	0.84	0.61	21.42
w/o SSL features	3.43	4.16	0.88	0.88	0.69	17.06	3.39	3.94	0.80	0.71	0.52	43.64
w/o codec-encoder features	3.40	4.22	0.91	0.94	0.72	6.76	3.38	4.17	0.88	0.91	0.66	11.29
CFM $\rightarrow$ regression refiner	3.42	4.25	0.95	0.97	0.90	3.61	3.43	4.24	0.90	0.93	0.77	9.11
trained on generated tokens	3.33	4.05	0.95	0.97	0.91	3.52	3.39	4.08	0.90	0.93	0.80	8.63
w/o z_q (no Stage 1)	3.34	4.04	0.95	0.97	0.91	4.08	3.36	4.08	0.90	0.93	0.80	9.22

sequential autoregressive decoding increases inference latency, and speaker similarity in complex scenarios still has room for improvement. Future work will consider improved speaker preservation and more efficient token generation.

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A. SPECTROGRAM COMPARISON ON DNS1 WITH-REVERB SAMPLES

Figures A1 to A4 show four utterances from the DNS1 *with-reverb* test set, one per figure. Each card contains the spectrogram of one system output (0–8 kHz, normalized to its own peak with an 80 dB range), the WER of its Whisper large-v3 transcript with the numbers of substituted (S), deleted (D), and inserted (I) words, and the transcript itself. The test set provides no transcripts, so the reference is the Whisper transcript of the clean (anechoic) recording, as in the dWER of section 3.3. Transcripts are shown after the text normalization used for scoring (lower-casing and removal of punctuation), and words that are substituted or inserted relative to the reference are marked in red.

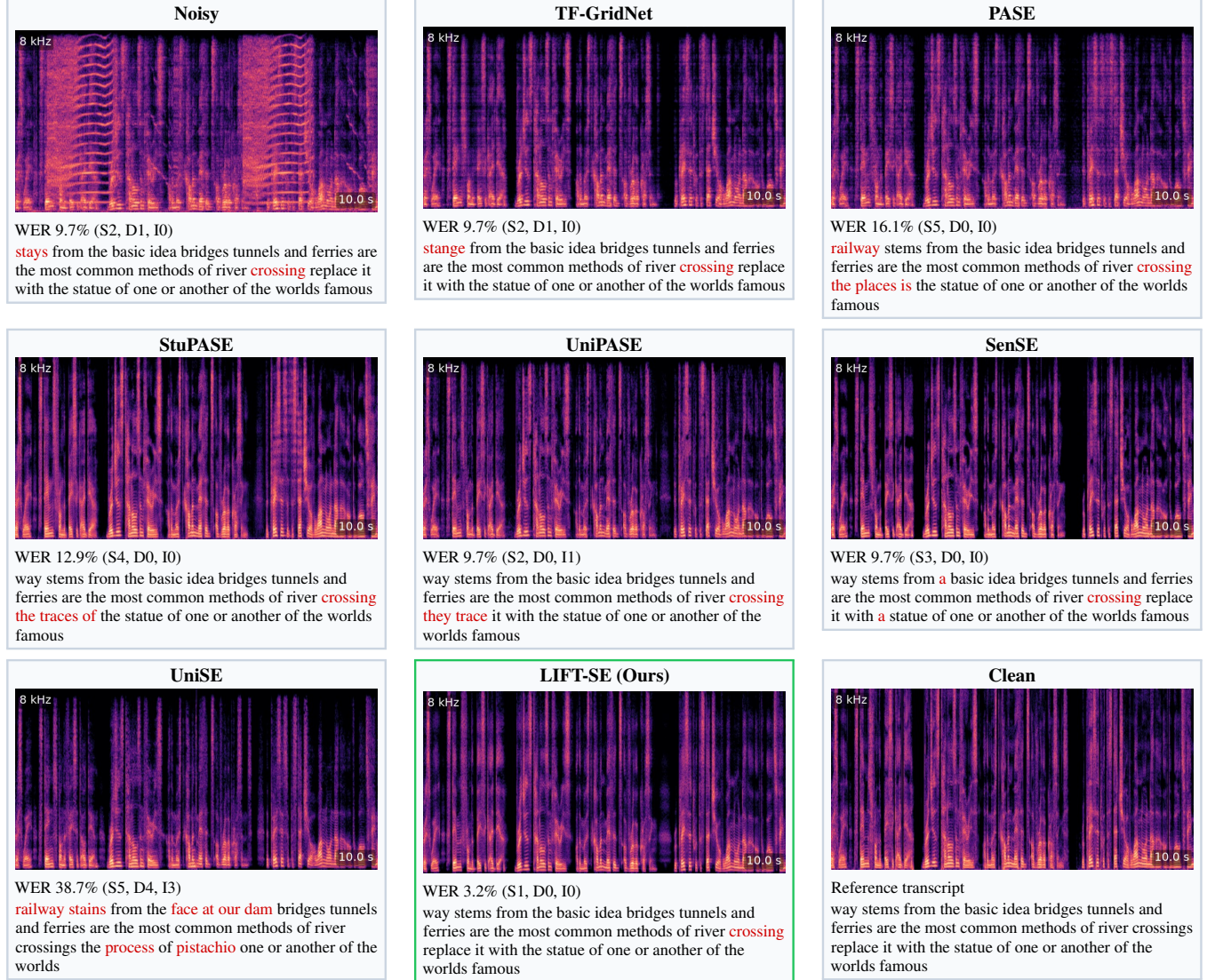


Fig. A1. Spectrograms and Whisper large-v3 transcripts of DNS1 *with-reverb* utterance fileid_76.



Fig. A2. Spectrograms and Whisper large-v3 transcripts of DNS1 *with-reverb* utterance fileid_96.

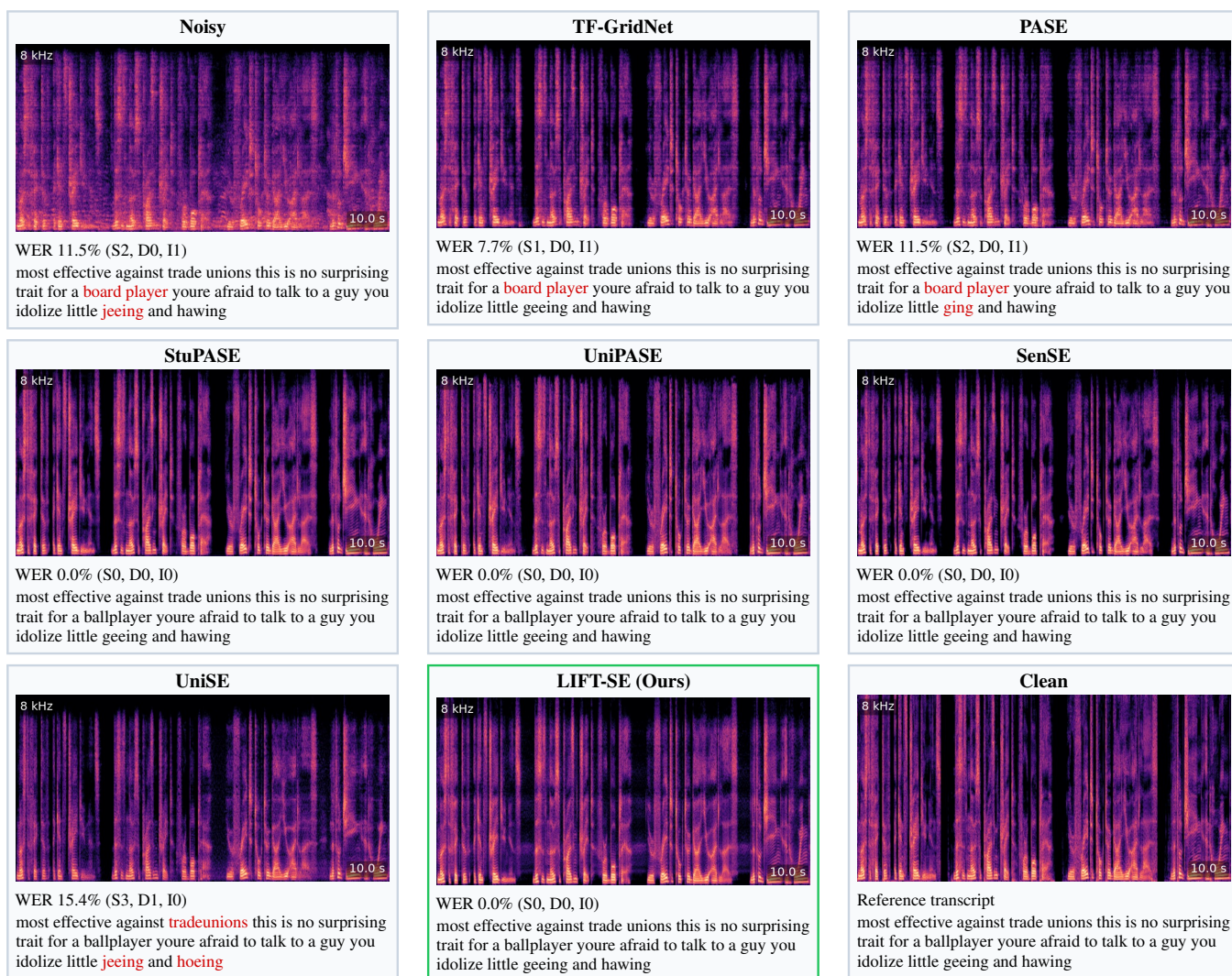


Fig. A3. Spectrograms and Whisper large-v3 transcripts of DNS1 *with-reverb* utterance fileid_21.



Fig. A4. Spectrograms and Whisper large-v3 transcripts of DNS1 *with-reverb* utterance fileid_73.